

YEONGWOL, Korea, Republic of

WMO: 471210

Lat: 37.1814N

Lon: 128.4575E

Elev: 793

StdP: 14.28

Time Zone: 9.00 (E09)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.3	(c) 10.0	(d) -8.9	(e) 3.5	(f) 16.0	(g) -5.1	(h) 4.3	(i) 17.5	(j) 14.5	(k) 25.9	(l) 12.3	(m) 27.5	(n) 2.1	(o) 270	(p) 0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.7	(c) 89.9	(d) 73.9	(e) 87.5	(f) 72.8	(g) 85.2	(h) 71.6	(i) 77.1	(j) 84.7	(k) 76.1	(l) 83.1	(m) 75.0	(n) 81.5	(o) 4.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	136.8	79.5	74.2	131.6	78.9	73.1	126.5	78.3	41.2	84.9	40.2	83.4	39.2	81.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.1	(b) 11.3	(c) 9.7	(e) 0.3	(f) 94.2	(g) 4.8	(h) 2.5	(i) -3.2	(j) 96.0	(k) -6.1	(l) 97.4	(m) -8.8	(n) 98.8	(o) -12.3	(p) 100.7		
(4) 13.1	(5) 11.3	(6) 9.7	(7) 0.3	(8) 94.2	(9) 4.8	(10) 2.5	(11) -3.2	(12) 96.0	(13) -6.1	(14) 97.4	(15) -8.8	(16) 98.8	(17) -12.3	(18) 100.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	25.8	30.8	40.8	52.4	62.5	70.6	75.8	76.6	67.7	56.3	42.5	29.7
	DBStd	18.69	7.43	7.10	6.72	6.70	4.74	4.24	3.97	4.66	5.09	6.27	7.56	7.42
	HDD50	2527	749	537	294	50	0	0	0	0	29	274	676	1095
	HDD65	5486	1214	957	750	379	104	8	0	0	29	274	676	1095
	CDD50	3531	0	0	9	123	389	619	801	824	531	216	19	0
	CDD65	1016	0	0	0	2	28	177	336	359	110	4	0	0
	CDH74	7547	0	0	0	66	464	1356	2353	2730	559	19	0	0
	CDH80	2231	0	0	0	7	82	370	723	957	92	0	0	0
(14) Wind	WSAvg	3.1	2.9	3.2	3.8	4.1	3.5	3.1	2.8	2.8	2.8	2.6	2.7	2.7
(15) Precipitation	PrecAvg	51.3	0.7	1.1	2.0	3.3	3.6	6.1	12.8	11.2	5.9	2.0	1.6	0.9
	PrecMax	80.7	2.0	2.8	5.7	7.6	8.2	17.0	33.7	26.3	14.8	5.9	5.0	2.2
	PrecMin	29.2	0.1	0.1	0.4	0.7	0.7	1.9	3.7	2.4	0.9	0.1	0.2	0.1
	PrecStd	12.3	0.6	0.8	1.3	1.9	1.8	3.5	6.3	6.6	4.2	1.5	1.1	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	55.7	68.0	80.4	86.1	89.6	92.4	94.1	86.4	76.5	66.5	51.5
		MCWB	39.0	43.9	51.5	58.3	62.5	67.1	75.9	75.7	71.7	61.9	54.7	43.5
	2%	DB	43.2	50.8	62.9	75.3	82.1	86.8	89.2	90.8	82.6	72.9	61.6	47.1
		MCWB	36.5	40.7	47.2	55.8	60.0	66.9	74.7	75.5	68.7	59.8	51.7	40.5
	5%	DB	40.3	46.8	58.4	71.4	79.1	84.3	86.7	88.2	79.9	70.0	58.3	43.7
		MCWB	34.3	37.7	45.1	53.1	59.4	66.3	73.9	75.0	67.5	58.8	50.4	38.1
	10%	DB	37.4	43.0	54.7	67.4	76.1	81.7	84.2	85.5	77.2	67.2	54.8	41.1
		MCWB	32.2	35.1	43.2	51.4	58.4	65.7	73.1	74.1	66.4	57.5	48.3	35.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	47.6	53.9	63.1	67.7	73.2	78.1	79.2	74.6	66.7	58.1	45.8
		MCDB	44.5	50.7	62.0	73.0	74.1	80.6	85.4	86.7	82.3	69.8	62.2	49.5
	2%	WB	37.7	41.7	50.4	58.5	64.7	71.4	77.0	77.5	72.6	63.4	54.5	41.7
		MCDB	41.6	49.1	58.9	69.8	73.9	79.0	83.9	85.6	78.4	68.7	59.2	45.7
	5%	WB	35.1	38.7	46.9	56.1	62.7	69.9	76.0	76.4	70.7	60.9	51.6	38.9
		MCDB	39.1	45.1	56.1	66.3	72.2	77.9	82.5	84.2	75.6	66.9	56.7	43.0
	10%	WB	32.8	36.0	44.0	53.6	61.0	68.4	75.0	75.5	68.8	58.7	49.1	36.4
		MCDB	36.8	42.2	53.3	63.8	71.1	76.2	81.1	82.6	74.1	65.4	53.9	40.3
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.9	20.8	23.5	23.3	19.8	13.9	14.7	17.1	20.5	19.1	17.6
		MCDBR	19.5	24.7	28.3	31.2	30.4	25.3	18.7	18.9	20.9	23.1	23.3	19.8
		MCWBR	13.2	15.4	15.0	13.5	10.6	7.8	5.8	5.9	7.8	10.2	13.4	13.5
		MCDBR	16.4	21.6	24.4	24.3	21.6	17.4	14.5	15.7	14.8	17.3	19.0	18.2
	5% WB	MCWBR	12.1	14.6	14.8	12.2	9.8	6.6	5.6	5.9	6.7	8.5	12.5	13.5
(40) Clear-Sky Solar Irradiance	taub		0.401	0.458	0.552	0.586	0.592	0.648	0.611	0.549	0.473	0.458	0.431	0.395
	taud		2.006	1.894	1.732	1.715	1.739	1.682	1.831	1.991	2.141	2.048	2.046	2.059
	Ebn at Noon		235	236	226	228	229	216	224	235	247	234	223	227
	Edh at Noon		42	53	68	73	73	77	66	55	45	45	40	37
(44) All-Sky Solar Radiation	RadAvg		807	1046	1331	1557	1756	1628	1320	1334	1231	1114	796	718
	RadStd		61	98	141	122	156	126	187	185	152	87	100	58

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.17	N/A	N/A	N/A	N/A	N/A	-227	-344	+254	+104
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+90

Nomenclature: See separate page